

CSE340: Computer Architecture

Quiz 2

Time: 40 minutes

Semester: Summer 26

Total Marks: 20

Name: _____ ID: _____ Section: _____

Question 1 [10]

Convert the corresponding code snippet to RISC-V assembly language.

```
if (A[i*2] == 0){  
    even = 4444;  
else{  
    odd = 1111;  
}
```

Assume A is a half-word array, the base address of A is in X18. Variables i, even, odd in X20, X21, X22.

Hard constraint: You can only use one shift operation in your code.

Answer:

```
slli x5, x20, 2  
add x5, x5, x18  
lh x5, 0(x5)  
bne x5, x0, else  
lui x21, 1  
addi x21, x21, 348  
beq x0, x0, exit  
else:  
addi x22, x0, 1111  
  
exit:
```

Question 2 [6]

If there is a U-type instruction in the previous code of your question, convert it to its corresponding machine code. Assume the opcode for U-type instruction is 1010110. Your answer must be in hexadecimal.

Answer:

Immediate	Rd	Opcode
0000 0000 0000 0000 0001	10101	1010110

Hex = 0x00001AD6

Question 3

Explain why we need U-type instructions. Then show how the U-type instruction changes the value of rd from the first question. [2+2]

Answer:

Addi can't add more than 12 bits of immediate.

x21 = 00....00 0000 0000 0001 0000 0000 0000

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Question 1 [10]

Convert the corresponding code snippet to RISC-V assembly language.

```
if (A[i*2] != 0){  
    odd = 5555;  
else{  
    even = 2222;  
}
```

Assume A is a word array, the base address of A is in X18. Variables i, even, odd in X20, X21, X22.

Hard constraint: You can only use one shift operation in your code and the destination register must be x5.

Answer:

```
slli x5, x20, 3  
add x5, x5, x18  
lh x5, 0(x5)  
beq x5, x0, else  
lui x22, 1  
addi x22, x22, 1459  
beq x0, x0, exit  
else:  
addi x21, x0, 2222  
  
exit:
```

Question 2 [6]

If there is a modified I-type instruction in the previous code of your question, convert it to its corresponding machine code. Assume the opcode, funct3, funct6 for I-type instruction is 1010100, 001, 000100. Your answer must be in hexadecimal.

Answer:

<u>funct6</u>	<u>immediate</u>	<u>rs1</u>	<u>funct3</u>	<u>rd</u>	<u>opcode</u>
0001 00	000011	10100	001	00101	1010100

Hex = 0x103A12D4

Question 3

- a. Explain why we need modified I-type instructions. [2]

Answer:

To ensure the maximum shifting is contained within 63.

- b. Show how the U-type instruction changes the value of rd from the first question. [2]

Answer:

x22 = 00...00 0000 0000 0000 0000 0001 0000 0000 0000

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Question 1 [10]

Convert the corresponding code snippet to RISC-V assembly language.

```
if (i == 0){  
    A[i*2] = 4444;  
} else {  
    i = i + 1;  
}
```

Assume A is a half-word array, the base address of A is in X18. Variable i in X20.

Hard constraint: You can only use one shift operation in your code. Assume rs1, rs2 of S type operation is x5, x6.

Answer:

```
Bne x20, x0, else  
Lui x6, 1  
Addi x6, x0, 348  
Slli x5, x20, 2  
Add x5, x5, x18  
Sd x6, 0(x5)  
beq x0, x0, exit  
else:  
addi x20, x20, 1  
  
exit:
```

Question 2 [6]

Convert the S-type instruction in the if block in the previous code of your question to its corresponding machine code. Assume the opcode, funct3 for S-type instruction is 1000110, 101. Your answer must be in hexadecimal.

Answer:

Immediate[5:11]	rs2	rs1	funct3	immediate[0:4]	opcode
0000000	00110	00101	101	00000	1000110

Hex: 0x0062D046

Note: in the question I wrote S type in else block however there is none in else block. Writing either the if block's S type as machine code or that there is no S type instruction in else block will get full marks.

Question 3

- a. Explain why S-type instructions have the immediate field separated.

Answer:

So that it matches the format with other instructions for easier decoding.

- b. Show how the U-type instruction changes the value of rd from the first question. [2]

Answer:

x6 = 00...00 0000 0000 0000 0000 0001 0000 0000 0000

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Question 1 [10]

Convert the corresponding code snippet to RISC-V assembly language.

```
if (i != 0){  
    i = i + 1;  
else{  
    A[i*2] = 5555;  
}
```

Assume A is a word array, the base address of A is in X18. Variable i in X20.

Hard constraint: You can only use one shift operation in your code. Assume rd of U-type operation is x5.

Answer:

```
beq x20, x0, else  
Addi x20, x20, 1  
Beq x0, x0, exit
```

```
else:  
lui x5, 1  
addi x5, x5, 1459  
Slli x6, x20, 3  
Add x6, x6, x18  
Sd x5, 0(x6)
```

exit:

Question 2 [6]

If there is a U-type instruction in the previous code of your question, convert it to its corresponding machine code. Assume the opcode for U-type instruction is 1010100. Your answer must be in hexadecimal.

Answer:

Immediate	Rd	Opcode
0000 0000 0000 0000 0001	00101	1010100

Hex = 0x000012D4

Question 3

- a. Explain why we need funct3 in I-type instructions. [2]

Answer:

Sign and size of data for load instructions.

- b. Show how the U-type instruction changes the value of rd from the first question. [2]

Answer:

x5 = 00...00 0000 0000 0000 0000 0001 0000 0000 0000